

NISHITH KUMAR GUPTA

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Education

Birla Institute of Technology and Science, Pilani

B.E in Mechanical Engineering, Minor in Computing and Intelligence

Hyderabad, Telangana

November 2020 - July 2024

Academic Heights Public School

CBSE, Percentage - 90.6

Gorakhpur, UP

April 2018 - August 2020

Relevant Coursework

- Data Structures
- Machine Learning
- Operating System
- Data Mining
- OOP
- Deep Learning
- Database Management
- NLP

Experience

Business Analyst – Integration & Data Management

Jun 2024 - Feb 2025

Veritas Prime

Hyderabad, Telangana

- Engineered and optimized end-to-end data integration solutions for **SAP SuccessFactors Employee Central**, leveraging **Python**, **Java**, **Groovy**, and **RESTful APIs** to automate global payroll data processing and built **10 +** scalable interfaces
- Designed robust data pipelines using **SQL** and **ETL** methodologies for complex transformation of enterprise-scale datasets
- Integrated **REST** and **SOAP** APIs to synchronize HRIS platforms with external systems, reducing manual interventions
- Deployed integration solutions in cloud environments, managing version control with **Git**, **CI/CD** pipelines, **Jira**, and automated testing frameworks to ensure minimal downtime and seamless production releases

Machine Learning Intern

Jul 2023 - Dec 2023

Distributed Energy

Pune, Maharashtra

- Developed an end-to-end **Machine Learning** model to forecast Solar Tariff/PPA prices, improving pricing accuracy by **15%**
- Fine-tuned ML algorithms (**Random Forest**, **XGBoost**, and **Neural Networks**), reducing **MAE** by **12%** from the baseline
- Built an automated **ETL** pipeline leveraging Pandas, NumPy, and **SQL** for data ingestion, preprocessing, and transformation
- Deployed the model using **Docker** and **GCP/AWS**, integrating a **Flask API** with an interactive web-based dashboard
- Major Tech Stack Used: Python, NumPy, Pandas, Flask, SQL, Docker, AWS/GCP

Projects

Calorie Burn Prediction | *ML Supervised Regression* **Kaggle**

May 2025

- Built a **supervised ML** pipeline using **ensemble techniques** (XGBoost, LightGBM, CatBoost) to predict calories burned from biometric data, securing a top **20% rank** out of **4300+** submissions in a Kaggle competition
- Engineered features (BMI, MET score, etc.) and optimized models with **K-Fold** cross-validation, achieving **0.058 RMSLE**
- Applied **SHAP** for model interpretability and feature importance analysis; leveraged **Optuna** for hyperparameter tuning to enhance model performance

Vendor Performance Analysis | *Python, SQL, Power BI* **GitHub**

Apr 2025 - May 2025

- Designed a comprehensive vendor performance analysis project using **SQL**, **Python** (Pandas, Seaborn, Matplotlib), and **Power BI** to process and extract insights from multiple datasets, including **1M+** raw retail transaction records
- Joined and filtered relevant datasets using **SQL**, performed **EDA**, correlation analysis, and **hypothesis testing** to uncover top vendors, inventory inefficiencies, and pricing anomalies driving strategic decisions
- Built a **Power BI** dashboard presenting metrics like vendor sales contribution, inventory turnover, and profit margin segmentation

Hospital Readmission Risk Classifier | *Python, LightGBM, Scikit-learn* **GitHub**

Oct 2024 - Dec 2024

- Built a machine learning model to predict hospital readmissions, addressing a critical healthcare challenge that costs **U.S.** hospitals over **\$41 billion** annually due to frequent readmissions within **30 days** of discharge
- Tested and optimized multiple machine learning algorithms, including **LightGBM (LGBM)**, **XGBoost**, and **Random Forest**, leveraging hyperparameter tuning and ensemble methods to enhance accuracy, with **LGBM** achieving the highest precision
- Performed **EDA**, uncovering critical insights into patient demographics & comorbidities to reduce readmissions & legal penalties

Paris 2024 Olympic Performance Dashboard | *Python, Power BI, Excel* **GitHub**

Sep 2024 - Oct 2024

- Automated dataset acquisition using the **Kaggle API**, with subsequent data cleaning and preprocessing using **Python (Pandas)**
- Built a **Power BI** dashboard with medal, athlete, and country-wise insights using slicers, filters, tooltips, and drill-downs
- Applied **Power Query** for ETL and developed advanced **DAX** measures for dynamic KPIs, time-based comparisons (e.g., YoY medal growth), and modeled sponsor-level ROI projections tied to country and athlete performance

Technical Skills

Languages: Python, Java, C/C++, SQL, Groovy, HTML/CSS

Developer Tools: AWS, Azure, Snowflake, Tableau, Power BI, GitHub, VS Code, MongoDB, MS Excel

Frameworks: TensorFlow, Apache Spark (PySpark), Flask, Django

Libraries: Pandas, NumPy, NLTK, Matplotlib, ScikitLearn

Statistical Techniques: Hypothesis Testing, A/B Testing, Regression Analysis, Root Cause Analysis

Certifications

Machine Learning Specialization | *Coursera*

- Supervised
- Unsupervised
- Reinforcement Learning